NAME: Diya Shah

MODULE: 8 ROUTING AND SWITICHING CONCEPTS

1. Explain Switch

A network switch is a device that connects multiple devices (PCs, printers, servers) in a LAN.

- It works at Layer 2 (Data Link Layer) of the OSI model
- It uses MAC addresses to forward data
- It sends data only to the intended device, not all devices (unlike hub)

2. Explain Switch Boot Sequence

When a Cisco switch is powered ON, it follows these steps:

1. POST (Power-On Self-Test)
    - Checks hardware (CPU, memory)
2. Boot Loader Execution
    - Stored in ROM, initializes system
3. Load IOS (Operating System)
    - IOS is loaded from Flash memory
4. Load Configuration File
    - Startup-config is loaded from NVRAM
5. Switch Ready
    - Switch becomes operational

3. Three Methods to Access Switch CLI

1. Console Access

- Direct connection using console cable
- Used for initial configuration

2. Telnet

- Remote access over network
- Not secure (plain text passwords)

3. SSH (Secure Shell)

- Remote access but encrypted (secure)
- Preferred method

4. Cisco IOS – Explanation & Configuration

Cisco IOS (Internetwork Operating System)

- The operating system used in Cisco devices
- Provides commands to configure and manage the switch

```
Switch> enable
Switch# configure terminal

Switch(config)# hostname SW1
SW1(config)# enable secret class

SW1(config)# line console 0
SW1(config-line)# password cisco
SW1(config-line)# login

SW1(config)# line vty 0 4
SW1(config-line)# password cisco
SW1(config-line)# login

SW1(config)# service password-encryption

SW1(config)# banner motd #Unauthorized access prohibited#

SW1(config)# end
SW1# copy running-config startup-config
```

5. Explain Switch Port

A switch port is the physical interface on a switch used to connect devices.

Types:

- Access Port
    - Connects to end devices
    - Belongs to one VLAN only

- Trunk Port
    - Connects switches/routers
    - Carries multiple VLANs

6. **R1, R2, R3, and R4 have their Fast Ethernet 0/0 interfaces attached to the same VLAN. A network engineer has typed a configuration for each router by using a word processor. He will later copy and paste the configuration into the routers. Examine the following exhibit, which lists configuration for the four routers, as typed by the network engineer. Assuming that all four routers can ping each other's LAN IP addresses after the configuration has been applied, choose the routers that will be able to form a neighbor relationship with the other routers on the LAN. (You can assume that, if not shown in the exhibit, all other related parameters are still set to their defaults.) (Choose two)**

From the image:

- All routers (R1, R2, R3, R4) are:
    - Connected to the same VLAN
    - Using OSPF
    - Advertising network 10.x.x.x
- Only difference:
    - router ospf 0
    - router ospf 1
    - router ospf 2
    - router ospf 3

NOTE: OSPF process ID is locally significant only
So even if process IDs are different, routers can still become neighbors

THEREFORE, All routers (R1, R2, R3, R4) will form neighbor relationships

7-enable secret [password] is hashed using the algorithm.

A. MD5
B. AH
C. PSK
D. ESP
E. WPA

8- An engineer connects to Router R1 and issues a show ip ospf neighbor command. The status of neighbor 2.2.2.2 lists FULL/BDR. What does the BDR mean?

A. R1 is an Area Border Router.
B. R1 is a backup designated router.
C. Router 2.2.2.2 is an Area Border Router.
D. Router 2.2.2.2 is a backup designated router.

9-Which command is used to view the neighbor discovery table on a PC?

A. show ipv6 neighbor
B. show ipv6 neighbors
C. netsh interface ipv6 show neighbor
D. netsh interface ipv6 show neighbors

10-What type of variable is being shown? Routers = [R1,R2,R3]

A. List

B. Dictionary

C. Simple

D. Unsigned integers

11- Identify the fields in an IPv4 header. (Choose three)

A. Host component

B. Time to Live

C. Source address

D. Destination address

E. Network